



Pedro Tarancón Álvarez

Theoretical Physicist — Holography & Machine Learning

Research Interests

My research lies at the interface of **theoretical high-energy physics** and **machine learning**. I study **strongly coupled matter and gravity** with applications to cosmology and astrophysics: the hydrodynamics of bubbles in **first-order phase transitions** with non-conformal equations of state, in the early universe and in **neutron star mergers**, and the resulting **gravitational-wave** signal. I use **gauge/gravity duality (AdS/CFT)** both to describe their far-from-equilibrium dynamics and to reconstruct the gravitational duals of strongly coupled quantum field theories from their equations of state. I am also interested in **turbulence and the geometry of fluid flows**. On the methods side, I develop **Physics-Informed Neural Networks (PINNs)** for the highly non-linear inverse and PDE problems arising in gravity, holography and mathematical physics, study the **latent representations** that networks learn for families of PDE solutions, and contribute to **AI agents for scientific discovery**.

Publications

- [1] D. Mateos, M. Sanchez-Garitaonandia and P. Tarancón-Álvarez, *Non-conformal obstructions to bubble expansion*, *arXiv e-prints* (2026) [2607.27874].
- [2] R. Jimenez, S. Mayboroda, P. Protopapas, L. Sàrieddine, D.N. Spergel and P. Tarancón-Álvarez, *Physics-Informed Neural Embeddings of PDE Solution Families*, *arXiv e-prints* (2026) [2607.06348].
- [3] R. Jimenez, D. Mateos, P. Protopapas, P. Solé-Vilaró, P. Tarancón-Álvarez and P. Tejerina-Pérez, *Gravitational Duals from Equations of State II: Large Hierarchies and False Vacua*, *JHEP* (accepted) (2026) [2606.30117].
- [4] A.A. Kalout, P. Tejerina-Pérez, K. Karchev, P. Tarancón-Álvarez, L. Sàrieddine, R. Jimenez et al., *Recovering Sharp Conductivity Features in the Finite-Data Calderón Problem with Physics-Informed Neural Networks*, *arXiv e-prints* (2026) [2606.28158].
- [5] P. Tarancón-Álvarez, L. Sàrieddine, P. Protopapas and R. Jimenez, *Learning embeddings of non-linear PDEs: the Burgers' equation*, *AI & PDE: ICLR 2026 Workshop on AI and Partial Differential Equations* (2026) [2603.07812].
- [6] F. Villaescusa-Navarro, B. Bolliet, P. Villanueva-Domingo, A.E. Bayer, P. Tarancón-Álvarez et al., *Denario Project: Deep Knowledge Artificial Intelligence Agents for Scientific Discovery*, *PRX Intelligence* **1** (2026) 021001 [2510.26887].
- [7] P. Tarancón-Álvarez, P. Tejerina-Pérez, R. Jimenez and P. Protopapas, *Efficient PINNs via multi-head unimodular regularization of the solutions space*, *Commun. Phys.* **8** (2025) 335 [2501.12116].
- [8] Y. Bea, R. Jimenez, D. Mateos, S. Liu, P. Protopapas, P. Tarancón-Álvarez et al., *Gravitational duals from equations of state*, *JHEP* **07** (2024) 087 [2403.14763].

Talks and Presentations

- 2026 *Non-conformal obstructions to bubble expansion* (talk, forthcoming, October), workshop “VAST 2026: From Vacuum Stability to Gravitational Waves”, Madrid, Spain.
- 2026 *Non-conformal obstructions to bubble expansion* (poster), Iberian Strings 2026, IGFAE, Santiago de Compostela, Spain.
- 2026 *Learning embeddings of non-linear PDEs: the Burgers equation* (poster), AI & PDE: ICLR 2026 Workshop, Rio de Janeiro, Brazil.
- 2025 *The Denario Project: Deep Knowledge Agents for Scientific Discovery* (oral), Cross-Collserola PhD Meeting, IFAE, Universitat Autònoma de Barcelona, Spain.
- 2025 *The Denario Project* (internal seminar, with L. Verde and R. Jimenez), ICCUB, University of Barcelona, Spain.
- 2025 *Gravitational Duals from Equations of State* (invited talk), workshop “New Insights in Black Hole Physics from Holography”, Instituto de Física Teórica (IFT), Madrid, Spain.
- 2025 *Gravitational Duals from Equations of State* (poster), conference “Gravity — New Quantum and String Perspectives”, Centro de Ciencias de Benasque Pedro Pascual, Benasque, Spain.
- 2025 *Gravitational Duals from Equations of State* (oral), Theoretical Physics Department, University of Florence, Italy.
- 2025 *Gravitational duals from equations of state* (gong show), Iberian Strings 2025, Bilbao, Spain.
- 2024 *Gravitational duals from equations of state* (seminar), Gran Sasso Science Institute, L'Aquila, Italy.

Teaching Experience

- 2025 – 2027 **Guest Lecturer — “Mathematical and Statistical Techniques”**, *Master’s in Astrophysics, Particle Physics and Cosmology, University of Barcelona*
Lecturing (in English) within the core methods course of the Master’s programme (42 students in 2025–26). Designed and delivered lectures and exercises on *Machine Learning, Physics-Informed Neural Networks and dimensionality reduction*; materials made available through the course GitHub repository.

Research Experience and Projects

- 2026 **Research visit (forthcoming)**, *Institute for Advanced Study, Princeton, USA*, Host: Prof. James M. Stone
Research stay (December 2026) on the effective description of vortex interactions and turbulence.
- 2023 – Present **PhD Researcher — Formación de Personal Investigador (FPI) Fellow**, *Institute of Cosmos Sciences (ICCUB), University of Barcelona*, Supervisors: David Mateos & Raúl Jimenez
Competitive Ministry of Science and Innovation fellowship for the doctoral thesis in Theoretical Physics and Astrophysics, within the project *Dark Energy and the Origin of the Universe*. Research on the holographic (AdS/CFT) description of strongly-coupled quantum fields in astrophysical and cosmological phenomena — cosmological phase transitions in the early universe and neutron-star mergers — using Physics-Informed Neural Networks to solve the associated non-linear inverse problems.
- 2025 **Research visit — Center for Computational Astrophysics, Flatiron Institute, New York, USA**, Host: Francisco Villaescusa-Navarro
Research visit (June 2025) working on the *Denario* multi-agent AI system.

- 2024 **Visiting Researcher — StellarDNN Lab**, *School of Engineering and Applied Sciences (SEAS), Harvard University*, Host: Prof. Pavlos Protopapas
Research stay (April–July 2024) developing Physics-Informed Neural Network methods for holography- and cosmology-related inverse problems. This collaboration is at the origin of an ongoing line of joint publications.
- 2022 – 2023 **Master's Thesis — Quantum chaos in de Sitter space and Centaur geometries**, *Instituto de Física Teórica UAM-CSIC (IFT) / Autonomous University of Madrid (UAM)*, Supervisors: Juan F. Pedraza & Ayan K. Patra, Grade: 9.8
Within the AdS/CFT framework, study of a 2d gravity solution describing a de Sitter interior smoothly glued to an anti-de Sitter exterior, diagnosing quantum chaos through out-of-time-ordered four-point functions computed from space-like geodesics anchored at the two AdS_2 boundaries.
- 2022 **Bachelor's Thesis — Structural relations of remote galaxies (CANDELS / 3D-HST)**, *Earth Physics and Astrophysics Department, Complutense University of Madrid (UCM)*, Supervisor: Jesús Gallego Maestro, Grade: 8.8
Analysis (in Python) of the CANDELS and 3D-HST catalogues to find structural relations between physical parameters of galaxies over the redshift range 0.4–1.5.
- 2021 – 2022 **Collaboration grantee (Ministry of Education)**, *Earth Physics and Astrophysics Department, Complutense University of Madrid (UCM)*
Development of a control prototype for the MOSAIC instrument of the ELT, within the GUAIX and ISCAR research groups (Profs. África Castillo and Jesús Gallego).

Education

- 2023 – 2027 **PhD in Theoretical Physics and Cosmology**, *Institute of Cosmos Sciences (ICCUB), University of Barcelona, Barcelona*
(exp.) Supervisors: David Mateos and Raúl Jimenez. Holographic principle (AdS/CFT) applied to strongly-interacting quantum fields in astrophysical phenomena, using Physics-Informed Neural Networks to solve inversion problems in highly non-linear systems of differential equations.
- 2022 – 2023 **MSc in Theoretical Physics (Particle Physics and Cosmology)**, *Autonomous University of Madrid (UAM)*, Grade: 9.33
Focus on the formal aspects of Quantum Field Theory, General Relativity and Quantum Gravity (Advanced QFT, Advanced Gravitation, Advanced Mathematics).
- 2018 – 2022 **BSc in Physics**, *Complutense University of Madrid (UCM)*, Grade: 9.02
Honors in Algebra (10), Scientific Computing Laboratory (9.7), Structure of Matter (9.8), Differential Geometry and Tensor Calculus (9.9), Observational Astronomy (9.5), Stellar Astrophysics (10) and Elementary Particles (9.9). Parts of the degree taught in English.

Fellowships and Awards

- 2023 – 2027 **FPI Predoctoral Fellowship**, Spanish Ministry of Science and Innovation (PRE2022-102220), awarded through national competition.
- 2021 – 2022 **Beca de Excelencia** (Excellence Scholarship), Comunidad de Madrid, for outstanding academic performance.
- 2021 – 2022 **Collaboration Grant**, Spanish Ministry of Education, for undergraduate research (UCM).

Professional Service

- 2026 – Organiser of the seminars of the Quantum Field Theory and String Theory group, ICCUB, present University of Barcelona.
- 2026 Referee for *Astronomy and Computing* (Elsevier).
- 2025 – Member of the **Simons Collaboration for the Geometry of Flows**, present

Software

SNOBEX (sole author): complete space of self-similar bubble solutions for an arbitrary equation of state, including shocked detonations.

NNHolo: PINN solver for the holographic inverse problem, used in both *Gravitational duals* papers.

Unimodular_regularization: multi-head PINN with unimodular regularization of the latent space.

PDE_embeddings (sole author): multi-head PINN framework and latent-space diagnostics for PDE solution families.

P2_ML_Lectures_public: lecture notes and exercises for the Master's course.

Contributor to AthenaK: implementation of turbulence forcing.

Schools, Workshops and Research Stays

- Workshop on Holography and Dense Matter, APC, Paris, France (2026).
- Geometry of Flows meeting, New York University, New York City, USA (2026).
- Geometry of Flows Simons Collaboration retreat — Explainability in ML & Multi-Agent Inference (mathematical hydrodynamics and turbulence), Canfranc Underground Laboratory, Spain (2026).
- Geometry of Flows Workshop (geometry of fluid flows, hydrodynamics and turbulence), Flatiron Institute, New York City, USA (2026).
- AI & PDE: ICLR 2026 Workshop on AI and Partial Differential Equations, Rio de Janeiro, Brazil (2026).
- IberianStrings 2026, IGFAE, Santiago de Compostela, Spain (2026).
- ETH-ITS Workshop “Geometry of Flows” (hydrodynamics and turbulence), ETH Zurich, Switzerland (2025).
- Simons Collaboration for the Geometry of Flows (mathematical hydrodynamics and turbulence), Institute for Advanced Study, Princeton, NJ, USA (2025).
- “Gravity — New Quantum and String Perspectives”, Centro de Ciencias de Benasque Pedro Pascual, Spain (2025).
- Workshop “New Insights in Black Hole Physics from Holography”, IFT Madrid, Spain (2025).
- GGI School on asymptotic symmetries and flat holography, Galileo Galilei Institute, Florence, Italy (2025).
- IberianStrings 2025, Bizkaia Aretoa, Bilbao, Spain (2025).
- Barcelona Initiative for Gravitation (BIG) Meetings (4th–7th), Barcelona, Spain (2024–2025).
- Cross-Collserola PhD Meeting in Astrophysics, Cosmology and Particles, Barcelona (2023, 2025).
- “The Complexity of the Cosmos” and “Gravitational duals from equations of state”, Gran Sasso Science Institute, L'Aquila, Italy (2024).
- COSMO CLASSIC 2024: future prospects for Cosmology, Asiago, Italy (2024).
- New Frontiers in Strong Gravity, Centro de Ciencias de Benasque Pedro Pascual, Spain (2024).
- StellarDNN Lab, SEAS, Harvard University — research stay with Prof. Pavlos Protopapas (Apr–Jul 2024).
- 2024 GGI School on “Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation”, Florence (60 h, 2024).
- CERN Winter School on Supergravity, Strings and Gauge Theory 2024, CERN, Geneva (2024).
- ComputeFest 2024: Physics-Informed Neural Networks (PINNs), Harvard IACS (16 h, 2024).
- ICCUB Machine Learning Workshop (2023) and ICCUB Winter Meeting (2024), Barcelona.

Technical Skills

Machine Learning	Physics-Informed Neural Networks (PINNs), deep learning for inverse and PDE problems, dimensionality reduction; PyTorch/JAX-based workflows and HPC clusters.
Programming	Advanced Python (numerical methods, data analysis, symbolic computation, ML) and Julia. Wolfram Mathematica (advanced). Matlab (intermediate). C++ (basic; implementing turbulence forcing in the <i>AthenaK</i> code). $\LaTeX$ (advanced).
Physics	AdS/CFT and holography, Quantum Field Theory, General Relativity, numerical relativity in asymptotically AdS spacetimes, relativistic hydrodynamics, cosmology.

Languages

Spanish	Mother tongue	
English	Fluent (working language of research and teaching)	<i>B2, EOI Raimundo de Toledo</i>
French	Basic	<i>A2, EOI Raimundo de Toledo</i>

Media and Outreach

- 2025 Research (lead author) featured in the Spanish national press — *La Vanguardia*, “UB researchers develop new AI techniques to solve complex equations in physics” (October 2025). [link]
- 2025 Video presentation “The Denario Project: Modular Automation of Scientific Research with Multi-Agent Systems”. [YouTube]

Additional Information

- Invited speaker, IX Conference on Introduction to Research, UCM, Madrid (2022).
- Amateur astronomy (telescope operation and calibration); piano (professional-level education, CPM Jacinto Guerrero); reading and music.